CUSTOMER CHURN ANALYSIS FOR TELECOM INDUSTRY

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1. Introduction

Customer churn is a critical issue in the telecom industry as losing customers directly impacts revenue and profitability. Retaining existing customers is more cost-effective than acquiring new ones. This project aims to analyze customer behavior, identify churn drivers, and build a predictive model to detect customers who are likely to leave.

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2. Abstract

The objective of this project is to analyze telecom customer data and predict churn using machine learning techniques. SQL was used for initial data analysis, and Python was used for data cleaning, visualization, and model building. A Logistic Regression model was developed to predict churn probability. Customers were segmented into risk categories, and business strategies were proposed to reduce churn.

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3. Tools Used

- SQL (Data Analysis)
- Python (Pandas, NumPy, Seaborn, Matplotlib)
- Scikit-learn (Logistic Regression)
- Jupyter Notebook

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4. Steps Involved in Building the Project

1. Imported dataset into SQL for analysis.
2. Performed churn rate and contract-type analysis.
3. Cleaned and preprocessed data in Python.
4. Converted categorical variables into numeric format.
5. Conducted Exploratory Data Analysis (EDA).
6. Built Logistic Regression classification model.
7. Evaluated model using accuracy, precision, recall.
8. Predicted churn probabilities.
9. Segmented customers into At Risk, Dormant, and Loyal groups.
10. Proposed business retention strategies.

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5. Results

The overall churn rate was approximately 26%.
Month-to-month contract customers showed highest churn probability.
Customers with low tenure and high monthly charges were more likely to churn.

The Logistic Regression model achieved an accuracy of 82.19%.

Customer Segmentation:

- **At Risk:** 6.05%
- **Dormant:** 23.90%
- **Loyal:** 70.05%

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6. Business Recommendations

For At Risk Customers:

- Provide discounts on monthly charges.
- Encourage conversion to annual contracts.
- Offer personalized retention calls.

For Dormant Customers:

- Launch targeted marketing campaigns.
- Provide bundled service offers.

For Loyal Customers:

- Introduce loyalty rewards.
- Offer referral incentives.

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7. Conclusion

This project successfully identified key churn factors and built a predictive model to classify customers based on churn risk. By implementing targeted retention strategies, telecom companies can reduce churn rate, improve customer satisfaction, and increase long-term profitability.